

The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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Abstract—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited large language model (LLM) and tool budgets. We introduce the Verified-Candidate Loop (VCL), which lets measured validation evidence control candidate promotion. A bounded quality-of-results score combines effective latency and worst-resource growth. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible measured evidence, while Failure Reflection turns tool diagnostics into the next constraint. On a balanced 150-task suite built from two fixed-commit AMD repositories, three hosted-model campaigns solve from 88.0% to 98.0% of tasks. Every model solves all 25 QoR-optimization tasks. Structural C/register-transfer-level (RTL) co-simulation repair accounts for the largest cross-model gap.

Index Terms—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

I. EVIDENCE MUST CONTROL LLM EDITS

HLS engineers must interpret unfamiliar C/C++ kernels, fix functional and streaming failures, and tune hardware under a small evaluation budget. Simulation and synthesis reports expose correctness and performance evidence [1]. An LLM can propose source changes, but its text cannot certify an interface, register-transfer-level (RTL) execution, or a hardware tradeoff.

LLM-assisted HLS systems automate code transformation and design-space exploration [2], [3]. General tool-using agents interleave model reasoning and actions [4]. A metered HLS loop also needs a candidate-promotion rule. An unchecked edit can pass compilation while failing a hidden test, deadlocking in RTL, missing 100 MHz, or consuming more hardware than its speedup justifies.

Our Verified-Candidate Loop (VCL) contributes three mechanisms. (1) A fail-closed interface and validation workflow preserves a verified fallback. (2) A bounded hardware score Q_{HW} rejects false speedups. (3) QoR-guided retrieval-augmented generation (QoR-RAG) and Failure Reflection focus later attempts on compatible evidence. We evaluate the complete agent in 450 hosted-model task runs.

II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface supplies one editable kernel, its top-function signature, public tests, target part, and budgets. The agent emits a complete source file and a structured report. Pre-flight rejects malformed inputs before the LLM or HLS tools consume budget. A tool server then records every candidate, command, gate, metric, token, and credit in one run state.

The VCL runs interface checks, C simulation (CSIM), HLS synthesis (SYNTH), the 100 MHz frequency check, the U55C capacity check, task-required C/RTL co-simulation (COSIM),

and metric-completeness checks. Let each applicable gate $G_i \in \{0, 1\}$. We set an inapplicable COSIM gate to one:

$$V(c) = G_{\text{int}} G_{\text{csim}} G_{\text{syn}} G_{\text{freq}} G_{\text{cap}} G_{\text{cosim}} G_{\text{metric}}. \quad (1)$$

Let b denote the verified fallback. The promotion rule is

$$P(c) = V(c) \wedge [Q_{\text{HW}}(c) > Q_{\text{HW}}(b)]. \quad (2)$$

Budget admission reserves the credits for all required gates. A failed gate or budget denial retains b . QoR-RAG supplies version-compatible rules, successes, and measured failures. Failure Reflection converts bounded diagnostics and the candidate diff into the next constraint without another LLM request. Hidden acceptance remains an evaluator-side check.

III. ONE SCORE REJECTS FALSE SPEEDUPS

The evaluator selects a valid starter as anchor a and falls back to a frozen reference that the agent cannot inspect. Let c denote a candidate, L its cycle latency, $p = \max(p_{\text{target}}, p_{\text{estimated}})$ its effective period, and II its initiation interval. The score uses the II branch only when the task marks II relevant and both values are positive:

$$r_L = \frac{p_a L_a}{p_c L_c}, \quad r_P = \begin{cases} r_L^{0.85} (II_a/II_c)^{0.15}, & \text{reliable } II, \\ r_L, & \text{latency only,} \end{cases} \quad (3)$$

For anchor and candidate resource counts A_j and C_j , let $\mathcal{R}$ contain lookup tables (LUTs), flip-flops (FFs), digital signal processing (DSP) blocks, block RAM (BRAM), and UltraRAM (URAM). The worst growth defines area quality:

$$\begin{aligned} \mathcal{R}_+ &= \{j \in \mathcal{R} : A_j > 1 \vee C_j > 1\}, \\ \widehat{\mathcal{R}} &= \begin{cases} \mathcal{R}_+, & \mathcal{R}_+ \neq \emptyset, \\ \mathcal{R}, & \mathcal{R}_+ = \emptyset, \end{cases} \\ r_A &= \left[\max_{j \in \widehat{\mathcal{R}}} \frac{\max(C_j, 1)}{\max(A_j, 1)} \right]^{-1}. \end{aligned} \quad (4)$$

The official score combines validity, bounded hardware utility, and budget efficiency:

$$h = r_P^{0.55} r_A^{0.45}, \quad Q_{\text{HW}} = 1 - \frac{1}{(1 + h)^2}, \quad (5)$$

$$\begin{aligned} E &= \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}}), \\ S &= 100VQ_{\text{HW}}E. \end{aligned} \quad (6)$$

Here the two u terms denote consumed fractions clipped to $[0, 1]$. Promotion compares Q_{HW} only after $V(c) = 1$,

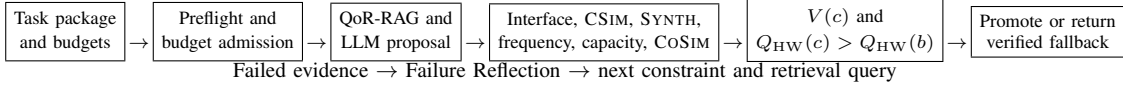


Fig. 1. **Measured evidence controls every state transition.** Only a valid hardware-score gain can replace the verified fallback.

TABLE I
STRUCTURAL REPAIR SEPARATES THE MODELS. ALL AND STRUCTURAL COLUMNS REPORT SUCCESS (%); QoR REPORTS THE MEAN OFFICIAL SCORE ON 25 OPTIMIZATION TASKS.

Model	All	Struct.	QoR	Tok. (M)
DeepSeek V4	98.0	96	80.37	3.77
Qwen3.5-122B	97.3	100	77.39	4.70
Qwen3.6-27B	88.0	52	76.80	4.27

so remaining budget cannot turn an inferior design into the fallback.

IV. THREE MODELS EXPOSE THE HARD CASES

Tasks and sources. The frozen suite contains 150 tasks and balances 25 tasks each across generation, compile repair, synthesis repair, functional repair, structural CoSIM repair, and QoR optimization. It derives 109 kernels from AMD’s Vitis HLS introductory examples and 41 from its acceleration examples at fixed commits [5], [6]. The builder adds controlled faults, public tests, evaluator-only hidden tests, and frozen references. Appendix C records the complete composition and provenance.

Protocol. Three campaigns call hosted models through application programming interfaces (APIs). They use the same 150 tasks, 9,000 credits, agent source revision, Alveo U55C target, and Vitis 2025.2. The manifest-defined gate contract applies the Track-A runtime requirements [7]. We count success only when the output passes CSIM, SYNTH, 100 MHz, capacity, and every task-applicable CoSIM gate. The 25 structural tasks require CoSIM. Missing or failed required evidence counts as failure. Reports record model identity, hidden-test outcome, tokens, tool calls, credits, frequency, and hardware metrics.

Results. DeepSeek V4 Pro leads with 147/150 successes, the highest QoR score (80.37), and 3.77 million tokens. Qwen3.5 solves 146 tasks with 4.70 million tokens. Qwen3.6 reaches only 52% on structural CoSIM repair, versus 96% and 100% for the other models. This category accounts for 12 of Qwen3.6’s 14-success gap to Qwen3.5. Every model solves all 25 QoR tasks, while their mean QoR scores still distinguish optimization quality.

Takeaway. Model choice changes structural repair reliability and token cost more than binary QoR-task completion.

V. EVIDENCE CONTROLS PROMOTION

The VCL makes validation evidence and Q_{HW} the promotion authority. The 450 runs show that models differ sharply on structural repair under this shared contract. Matched ablations and repeated campaigns must establish the causal effects of QoR-RAG and Failure Reflection. Broader repositories, tool

versions, and devices will test transfer beyond the present AMD corpus.

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APPENDIX A ONE STATE MAKES RUNS AUDITABLE

The command-line interface accepts a task package and an output root. The package exposes only the editable source, public testbench, top-function contract, target configuration, and budget. The agent returns a complete kernel and a machine-readable report. The evaluator keeps hidden tests and frozen references outside the agent-visible filesystem.

APPENDIX B SCORING HANDLES MISSING EVIDENCE

The production scorer floors each resource count at one before it computes a growth ratio. This rule avoids division by zero without rewarding an unused resource. It uses the weighted II term only when the task marks II relevant and both the anchor and candidate report positive values. The evaluator picks a valid starter as the anchor and falls back to a frozen evaluator-only reference. A missing anchor produces no official QoR score, but the task stays in the success-rate denominator. Candidate promotion compares Q_{HW} after validity and excludes the budget multiplier E .

APPENDIX C THE SUITE BALANCES SIX CAPABILITIES

The benchmark builder freezes every upstream path, commit, task description, public test, hidden test, reference, and task configuration in the accepted manifest. The corpus contains no cross-category kernel overlap. Before the campaigns, the builder quarantined 31 construction candidates that failed its acceptance checks and replaced them with validated tasks. The initial gate matrix accepts all 150 final packages.

APPENDIX D OUTPUTS ENFORCE APPLICABLE GATES

Table V scopes the claim of strict compliance to generated kernels and experimental reports. The final submission checklist separately tracks the PDF, supplemental source package, and demonstration video.

The paper source uses the IEEE conference class and starts these appendices after an explicit page break. Because this revision intentionally performs no local PDF build, the authors must confirm the two-page main-body boundary in the final compiled PDF.

APPENDIX E CAMPAIGNS SHARE ONE CONTRACT

APPENDIX F ONLY PROPOSAL CONSUMES MODEL TOKENS

APPENDIX G STRUCTURAL REPAIR SEPARATES MODELS

DeepSeek fails three tasks, Qwen3.5 fails four, and Qwen3.6 fails eighteen. Qwen3.6’s structural category contributes twelve more failures than Qwen3.5’s and explains most of

their aggregate difference. The three models all complete the QoR category, so binary success alone hides optimization quality. Their mean official QoR-category scores of 80.37, 77.39, and 76.80 provide that separation. The per-task reports retain the gate and diagnostic fields needed to subdivide future failures into interface, functional, synthesis, timing, capacity, structural, and budget causes.

APPENDIX H GENERATED MACROS KEEP RESULTS CURRENT

The main table and Appendix Tables IX and X read tracked macros from `technical-paper/results_generated.tex`. After replacing or extending the run directories, execute `python3 technical-paper/scripts/update_results.py` from the repository root. The script selects the latest matching final report, validates its model and task count, and refreshes all aggregate tables.

Future per-task exports will use one stable schema: task and category, candidate lineage, applicable gates, hidden acceptance, latency, II , period, five resources, Q_{HW} , official score, prompt and completion tokens, requests, tool calls, credits, wall time, status, and evidence hashes. This schema lets a new campaign replace aggregate rows without changing the paper narrative.

APPENDIX I CURRENT EVIDENCE LEAVES CAUSAL GAPS

Each model contributes one campaign, so provider variation and sampling noise remain unmeasured. The controlled-fault benchmark supports capability isolation but covers only two AMD repositories. Only the structural category requires COSIM. Tasks without a valid anchor contribute to success and receive no scored-mean entry. Matched seeds, repeated campaigns, broader repositories, newer tool releases, other FPGA devices, and the planned ablations will test robustness and causal mechanism effects.

APPENDIX J REPORTS PRESERVE REPRODUCTION EVIDENCE

Evaluators build the supplied Docker image, configure the model endpoint through environment variables, and mount the task and output directories. The entry point writes the final kernel, structured run report, and evidence under the selected output root. The Vitis child environment excludes credentials, and report writers redact sensitive text.

The repository includes the accepted manifest, public task materials, run reports, result-generation script, and paper source. Evaluator-only hidden tests and references remain inaccessible to the agent by design. The study uses no human participants or personal data. Authors must complete the funding, conflict-of-interest, contribution, and writing-assistance disclosures on the submission form without breaking double-blind review.

TABLE II
ONE RUN STATE CONNECTS PROPOSALS TO AUDITABLE EVIDENCE. THE OUTPUT WRITER PRESERVES THE LATEST CANDIDATE THAT PASSED EVERY REQUIRED GATE.

State group	Recorded fields
Task contract	Task identifier, editable path, top signature, target part, clock target, required gates, and credit limit
Candidate evidence	Source fingerprint, parent candidate, change summary, gate outcomes, diagnostics, evidence identifiers, and terminal status
Hardware evidence	Cycle latency, initiation interval, estimated period, LUT, FF, DSP, BRAM, URAM, Q_{HW} , and official score
Accounting	Prompt, completion, and total tokens; LLM requests; tool calls; credits; wall time
Output provenance	Model endpoint, agent source hash, manifest hash, report hash, and final kernel hash

TABLE III
THE SUITE BALANCES SIX DISTINCT AGENT CAPABILITIES. ONLY STRUCTURAL REPAIR REQUIRES CoSIM BY CONSTRUCTION.

Task type	Initial condition	Agent objective	Count	CoSIM required
Code generation	Signature-only stub	Generate kernel	25	0
Compile repair	Compile failure	Restore compilation	25	0
Synthesis repair	Synthesis failure	Restore synthesis	25	0
Functional repair	CSIM failure	Restore correctness	25	0
Structural repair	CoSIM failure	Restore RTL execution	25	25
QoR optimization	Valid, unoptimized	Improve hardware score	25	0

TABLE IV
FIXED COMMITS MAKE THE SOURCE CORPUS REPRODUCIBLE.

Repository	License	Upstream commit	Tasks
Xilinx/Vitis-HLS-Introductory-Examples	Apache-2.0	aa5c160faf5d5ebf58674df8f0591f9984ebae0f	109
Xilinx/Vitis_Accel_Examples	MIT	81187602355a7c2b666351154c5acca2074cae64	41

TABLE V
THE RUNTIME OUTPUT CONTRACT IMPLEMENTS EVERY APPLICABLE TRACK-A EVALUATION REQUIREMENT. MISSING EVIDENCE FAILS CLOSED.

Track-A requirement	Implementation and recorded evidence
Alveo U55C and Vitis 2025.2	The container fixes the target part and tool release. Each report records the execution-source hash.
CSIM, SYNTH, and CoSIM	Every task requires CSIM and SYNTH. The 25 structural tasks also require CoSIM. Any failed or missing applicable gate rejects the output.
At least 100 MHz	The frequency gate compares the achieved estimate with the 100 MHz floor and rejects a miss.
Resource feasibility and QoR	Capacity checks precede promotion. Reports store latency, II , clock, five resource counts, Q_{HW} , and the official score when an anchor exists.
Token consumption	Task and campaign reports store prompt, completion, and total tokens.
Three recommended model families	The campaigns use real-API endpoints for DeepSeek V4 Pro, Qwen3.5-72B-A10B, and Qwen3.6-72B. Every task record proves its selected endpoint.
Hidden benchmarks	The evaluator alone loads hidden tests and frozen references. The agent receives public files only.
Docker and reproducibility	Reports store model, kernel, manifest, report, and execution-source provenance. No campaign used a mock, scripted, or replay backend.

TABLE VI
ALL THREE CAMPAIGNS SHARE THE SAME EVALUATION CONTRACT.

Field	Fixed value
Hardware and tools	AMD Alveo U55C, Vitis 2025.2
Clock floor	100 MHz
Accepted corpus	tasks/track_a_150/candidate_manifest.json, 150 tasks
Tasks per campaign	150
Campaign budget	9,000 credits
Backend policy	Real API only; no mock, script, or replay
Execution source hash	e035c4b38859b53b9aa293de3a0a7bef5a09f43bf9a0b4f907387693000b0f86

TABLE VII
ONLY CANDIDATE PROPOSAL CONSUMES LLM TOKENS. TOOL PHASES CONSUME CREDITS AND WALL TIME UNDER SEPARATE ACCOUNTING.

Workflow phase	LLM token consumption	Recorded non-token cost
Task preflight and budget admission	0	Validation status
QoR-RAG query and context assembly	0	Retrieved evidence identifiers
Candidate generation, repair, and optimization	Prompt and completion tokens per request	LLM request count
Interface and HLS validation	0	Tool calls, credits, wall time
Failure Reflection and promotion	0	Failure signature and decision

TABLE VIII
CAMPAIGN REPORTS ACCOUNT FOR EVERY MODEL RESPONSE. VALUES USE MILLIONS OF TOKENS.

Model	Prompt	Completion	Total	API requests
DeepSeek V4 Pro	2.43	1.34	3.77	464
Qwen3.5-122B-A10B	2.05	2.64	4.70	403
Qwen3.6-27B	2.51	1.76	4.27	489

TABLE IX
DEEPSEEK LEADS AGGREGATE SCORE AND TOKEN EFFICIENCY. ALL-TASK MEAN ASSIGNS ZERO TO TASKS WITHOUT AN OFFICIAL SCORE; SCORED MEAN EXCLUDES TASKS THAT LACK A VALID QoR ANCHOR.

Model	Success	All-task mean	Scored mean	Tokens (M)	Credits	Calls
DeepSeek V4 Pro	147/150	54.27	70.78	3.77	2,823	791
Qwen3.5-122B-A10B	146/150	53.37	69.61	4.70	2,969	846
Qwen3.6-27B	132/150	47.38	71.07	4.27	2,358	691

TABLE X
STRUCTURAL CoSim REPAIR CAUSES THE MAIN SUCCESS GAP. ENTRIES REPORT SUCCESS RATE (%).

Model	Generate	Compile	Synth	Functional	Structural	QoR
DeepSeek V4 Pro	92	100	100	100	96	100
Qwen3.5-122B-A10B	88	100	100	96	100	100
Qwen3.6-27B	92	100	92	92	52	100

TABLE XI
MATCHED RERUNS WILL ISOLATE EACH EVIDENCE-REUSE MECHANISM. PENDING ROWS REMAIN OUTSIDE CURRENT CAUSAL CLAIMS.

Variant	Controlled change	Success	Repeat failures	Tokens	Credits	Mean Q_{HW}
Full agent	None	Pending	Pending	Pending	Pending	Pending
No QoR-RAG	Disable all retrieval	Pending	Pending	Pending	Pending	Pending
No failure cases	Retrieve rules and successes only	Pending	Pending	Pending	Pending	Pending
No Reflection	Remove diagnostic-to-constraint feedback	Pending	Pending	Pending	Pending	Pending